NEET UG 2027: Physics Formula Compendium

Module 04: Ray Optics, Wave Optics, Modern Physics & Semiconductors

Prepared, Compiled & Authored by Satya Prakash | **Subject:** Physics (Units 16, 17, 18 & 19)

Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

1 Ray Optics & Optical Instruments (Unit 16)

1.1 Spherical Mirrors, Snell's Law & Critical Angle

Mirror Formula, Magnification & Normal Shift

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R}$$ [Cartesian Sign Convention: Pole is origin; light dire (1)

$$m=-\frac{v}{u}=\frac{f}{f-u}=\frac{f-v}{f}$$ [Transverse / Linear Magnification] (2)

$$m_L=\frac{dv}{du}=-m^2=-\left(\frac{v}{u}\right)^2$$ [Longitudinal Magnification for small object along a (3)

$$\mu_1\sin i=\mu_2\sin r \implies \frac{\sin i}{\sin r}=\frac{\mu_2}{\mu_1}=\frac{v_1}{v_2}=\frac{\lambda_1}{\lambda_2}$$ [Snell's Law of Refraction; Frequency ν remains inva (4)

$$d_{\text{apparent}}=\frac{d_{\text{real}}}{\mu_{\text{relative}}}=d\left(\frac{\mu_{\text{observer}}}{\mu_{\text{object}}}\right), \qquad \Delta t=t\left(1-\frac{1}{\mu}\right)$$ [Apparent Depth & Normal Shift through slab] (5)

Total Internal Reflection (TIR) & Underwater Illuminance Cone:

$$\sin i_c=\frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}}=\frac{1}{\mu}, \qquad r_{\text{circle}}=\frac{h}{\sqrt{\mu^2-1}}, \qquad \text{Area}=\frac{\pi h^2}{\mu^2-1}$$ [Point source at depth h] (6)

1.2 Prisms, Dispersion & Spherical Refracting Surfaces

Prism Deviation, Dispersive Power & Curved Surface Refraction

$$\delta = (i_1 + i_2) - A, \qquad r_1 + r_2 = A$$ [General Triangular Prism of Angle A] (7)

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \iff i_1 = i_2 = \frac{A+\delta_m}{2}, \quad r_1 = r_2 = \frac{A}{2}$$ [Minimum Deviation Condition; ray para (8)

$$\delta_{\text{thin}} = (\mu - 1)A, \qquad \theta_{\text{dispersion}} = (\mu_V - \mu_R)A, \qquad \omega = \frac{\mu_V - \mu_R}{\mu_Y - 1}$$ [Thin Prism ($A \leq 10^\circ$) & Dispersive Pow (9)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}, \qquad m = \frac{\mu_1 v}{\mu_2 u}$$ [Refraction at Single Spherical Interface] (10)

1.3 Thin Lenses, Combinations & Silvered Systems

Lens Maker's Equation, Power & Silvering Formulations

$$\frac{1}{f} = \left(\frac{\mu_{\text{lens}}}{\mu_{\text{medium}}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$ [Lens Maker's Formula] (11)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}, \qquad m = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$ [Thin Lens Equation & Magnifica (12)

$$P = \frac{1}{f\ (\text{in m})} = \frac{100}{f\ (\text{in cm})} \quad [\text{D} = \text{m}^{-1}], \qquad P_{\text{contact}} = \sum P_i = P_1 + P_2$$ (13)

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \implies P = P_1 + P_2 - dP_1P_2$$ [Two Lenses Separated by Distan (14)

Silvering of a Lens Surface (Acts as an Equivalent Spherical Mirror):

$$P_{\text{eq}} = 2P_{\text{lens}} + P_{\text{mirror}} = \frac{2}{f_L} + \frac{1}{f_M} \implies F_{\text{eq}} = -\frac{1}{P_{\text{eq}}}$$ [System always acts as a mirror] (15)

Lens Cutting Rules & Medium Immersion Effects

- **Vertical Cut (Perpendicular to Principal Axis):** Cutting a convex lens of focal length f into two halves produces two plano-convex lenses: $f' = 2f, P' = P/2$. Intensity remains **unchanged**.
- **Horizontal Cut (Along Principal Axis):** Each half retains identical curvature: $f' = f, P' = P$. Intensity is halved because light-gathering aperture area is reduced by 50%.
- **Immersion in Liquid (μ_{med}):**

$$\frac{f_{\text{liquid}}}{f_{\text{air}}} = \frac{(\mu_g - 1)}{\left(\frac{\mu_g}{\mu_l} - 1\right)} \implies \text{For glass } (\mu = 1.5) \text{ in water } (\mu = 4/3): \quad f_{\text{water}} = 4f_{\text{air}}$$

If $\mu_l > \mu_g$, focal length reverses sign: a converging convex lens behaves as a diverging concave lens!

1.4 Optical Instruments

Instrument	Normal Adjustment ($v = \infty$)	Near Point Adjustment ($v = D = 25\text{cm}$)	Tube Length (L)
Simple Microscope	$m = \frac{D}{f}$	$m = 1 + \frac{D}{f}$	Eye at lens
Compound Microscope	$m = -\left(\frac{v_o}{u_o}\right)\left(\frac{D}{f_e}\right) \approx -\frac{L}{f_o}\frac{D}{f_e}$	$m = -\left(\frac{v_o}{u_o}\right)\left(1 + \frac{D}{f_e}\right) \approx -\frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$	$L = v_o + u_e \approx v_o + f_e$
Astronomical Telescope	$m = -\frac{f_o}{f_e}$	$m = -\frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$	$L = f_o + f_e$ or $f_o + u_e$

Table 1: Magnifying Powers and Tube Lengths of Canonical Optical Instruments.

2 Wave Optics (Unit 16)

2.1 Interference & Young's Double Slit Experiment (YDSE)

Interference Intensity, Fringe Position & Fringe Width

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi \implies I = 4I_0\cos^2\left(\frac{\phi}{2}\right) \quad (\text{when } I_1 = I_2 = I_0) \tag{16}$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2 = \left(\frac{a_1+a_2}{a_1-a_2}\right)^2, \qquad \Delta\phi = \frac{2\pi}{\lambda}\Delta x \tag{17}$$

$$y_n = n\frac{\lambda D}{d} \quad (n = 0, 1, 2, \dots) \qquad [n\text{-th Bright Fringe / Maximum;} \tag{18}$$

$$y'_n = (2n-1)\frac{\lambda D}{2d} \quad (n = 1, 2, 3, \dots) \qquad [n\text{-th Dark Fringe / Minimum; } \Delta \tag{19}$$

$$\beta = \frac{\lambda D}{d}, \qquad \theta = \frac{\beta}{D} = \frac{\lambda}{d} \quad [\text{rad}] \qquad [\text{Linear Fringe Width } \beta \text{ \& Angula} \tag{20}$$

Shift Produced by Thin Dielectric Slab of Thickness t and Refractive Index μ:

$$\Delta y = \frac{D}{d}(\mu - 1)t, \qquad N_{\text{shifted}} = \frac{\Delta y}{\beta} = \frac{(\mu-1)t}{\lambda} \tag{21}$$

2.2 Diffraction due to a Single Slit

Single Slit Diffraction Maxima, Minima & Fresnel Distance

$$a\sin\theta = n\lambda \implies y_n = n\frac{\lambda D}{a} \quad (n = 1, 2, 3, \dots) \qquad [\text{Diffraction Minima Condition}] \tag{22}$$

$$a\sin\theta \approx (2n+1)\frac{\lambda}{2} \implies y'_n = (2n+1)\frac{\lambda D}{2a} \qquad [\text{Secondary Maxima Condition } (n \geq 1)] \tag{23}$$

$$2\theta_0 = \frac{2\lambda}{a}, \qquad W_{\text{central}} = 2y_1 = \frac{2\lambda D}{a} = 2\beta_{\text{diffraction}} \qquad [\text{Width of Central Diffraction Maximum}] \tag{24}$$

$$Z_F = \frac{a^2}{\lambda} \qquad [\text{Fresnel Distance: Ray optics valid for } z \ll Z_F] \tag{25}$$

Wave Optics Traps: Interference vs Diffraction Central Fringe

- **Central Maximum Width in Diffraction is Doubled:** In single-slit diffraction, the central maximum spreads between the first minima on both sides ($-\lambda/a$ to $+\lambda/a$). Its linear width is $\frac{2\lambda D}{a}$, exactly **twice** the width of secondary maxima ($\frac{\lambda D}{a}$). In YDSE, all fringes have identical width $\beta = \frac{\lambda D}{d}$.
- **YDSE Immersed in Liquid:** When apparatus is dipped in liquid of refractive index μ, wavelength shrinks ($\lambda' = \lambda/\mu$). Hence fringe width decreases: $\beta' = \beta/\mu$. Angular width also shrinks: $\theta' = \theta/\mu$.

3 Dual Nature of Matter & Radiation (Unit 17)

3.1 Photoelectric Effect & Einstein's Equation

Einstein's Photoelectric Governing Equations

$$E = h\nu = \Phi_0 + K_{\max} = \Phi_0 + eV_0 \quad \text{[Einstein's Photoelectric Law]} \tag{26}$$

$$\Phi_0 = h\nu_0 = \frac{hc}{\lambda_0} \approx \frac{1240}{\lambda_0 \text{ (in nm)}} \text{ eV} \quad \text{[Work Function } \Phi_0\text{; Cutoff wavelength } \lambda_0\text{]} \tag{27}$$

$$eV_0 = h\nu - \Phi_0 \implies V_0 = \left(\frac{h}{e}\right)\nu - \frac{\Phi_0}{e} \quad \text{[Slope of } V_0 - \nu \text{ graph is universal: Slope} = h/e\text{]} \tag{28}$$

$$K_{\max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right) = 1240\left(\frac{1}{\lambda \text{ [nm]}} - \frac{1}{\lambda_0 \text{ [nm]}}\right) \text{ eV} \tag{29}$$

Photon Dynamics (Rest Mass $m_0 = 0$):

$$E = h\nu = \frac{hc}{\lambda}, \qquad p = \frac{E}{c} = \frac{h}{\lambda}, \qquad m_{\text{effective}} = \frac{h\nu}{c^2}, \qquad N_{\text{photons/sec}} = \frac{P}{h\nu} = \frac{P\lambda}{hc} \tag{30}$$

3.2 de Broglie Matter Wave Formulations

de Broglie Wavelength of Particles & Accelerating Potentials

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} \quad \text{[Fundamental Matter Wave Equation]} \tag{31}$$

$$\lambda_{\text{electron}} = \frac{h}{\sqrt{2m_e eV}} = \sqrt{\frac{150}{V}}\ \text{Å} = \frac{1.227}{\sqrt{V \text{ [Volts]}}}\ \text{nm} \quad [V = \text{accelerating potential in Volts}] \tag{32}$$

$$\lambda_{\text{proton}} = \frac{0.286}{\sqrt{V}}\ \text{Å}, \qquad \lambda_{\alpha} = \frac{0.101}{\sqrt{V}}\ \text{Å}, \qquad \lambda_{\text{deuteron}} = \frac{0.202}{\sqrt{V}}\ \text{Å} \tag{33}$$

$$\lambda_{\text{gas molecule}} = \frac{h}{\sqrt{3mk_BT}} = \frac{h}{\sqrt{2m\left(\frac{3}{2}k_BT\right)}} \quad \text{[Thermal de Broglie Wavelength at Temp } T\text{]} \tag{34}$$

Photoelectric Traps: Intensity vs Frequency Effects

- **Intensity Modifies Current, NOT Kinetic Energy:** Doubling the intensity of incident light doubles the number of photons per second, thereby doubling the **saturation photoelectric current** ($I_{\text{sat}} \propto$ Intensity). The maximum kinetic energy $K_{\max}$ and stopping potential V_0 remain **strictly unchanged**.
- **Frequency Modifies Kinetic Energy, NOT Current:** Increasing frequency ν increases $K_{\max}$ and V_0 linearly ($V_0 = \frac{h}{e}\nu - \frac{\Phi_0}{e}$). Saturation current is independent of frequency above ν_0.

4 Atoms & Nuclei (Unit 18)

4.1 Bohr's Hydrogen-Like Atom Model

Bohr Postulates, Radius, Velocity, Energy & Rydberg Formula

$$L = mv_n r_n = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots) \qquad \text{[Bohr Angular Momentum Quantization]} \tag{35}$$

$$r_n = \frac{n^2 h^2 \varepsilon_0}{\pi m Z e^2} = 0.529 \left(\frac{n^2}{Z}\right) \text{ Å} = r_1 \left(\frac{n^2}{Z}\right) \qquad \left[\text{Orbital Radius: } r \propto \frac{n^2}{Z}\right] \tag{36}$$

$$v_n = \frac{Ze^2}{2\varepsilon_0 nh} = \left(\frac{c}{137}\right)\frac{Z}{n} \approx 2.18 \times 10^6 \left(\frac{Z}{n}\right) \text{ m s}^{-1} \qquad \left[\text{Orbital Velocity: } v \propto \frac{Z}{n}\right] \tag{37}$$

$$E_n = -\frac{mZ^2 e^4}{8\varepsilon_0^2 n^2 h^2} = -13.6 \left(\frac{Z^2}{n^2}\right) \text{ eV} \qquad \left[\text{Total Energy: } K = -E_n, \quad U = 2E_n\right] \tag{38}$$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \qquad \left[R = 1.097 \times 10^7 \text{ m}^{-1}, \quad \frac{1}{R} \approx 912 \text{ Å}\right] \tag{39}$$

Total Number of Emission Lines from Excited State n to Ground State ($n = 1$):

$$N = \frac{n(n-1)}{2}, \qquad N_{\text{between } n_2 \text{ and } n_1} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} \tag{40}$$

Spectral Series	Lower Level (n_1)	Upper Level (n_2)	Spectral Region	Shortest Wavelength ($\lambda_{\min}$)
Lyman Series	$n_1 = 1$	$2, 3, 4, \dots$	Ultraviolet (UV)	$1/R \approx 912$ Å
Balmer Series	$n_1 = 2$	$3, 4, 5, \dots$	Visible & Near UV	$4/R \approx 3646$ Å
Paschen Series	$n_1 = 3$	$4, 5, 6, \dots$	Infrared (IR)	$9/R \approx 8205$ Å
Brackett Series	$n_1 = 4$	$5, 6, 7, \dots$	Infrared (IR)	$16/R \approx 14586$ Å
Pfund Series	$n_1 = 5$	$6, 7, 8, \dots$	Far Infrared	$25/R \approx 22790$ Å

Table 2: Hydrogen Spectral Series Classifications and Spectral Regimes.

4.2 Nuclear Size, Mass Defect & Binding Energy

Nuclear Radius, Density, Mass Defect & Fission/Fusion Energies

$$R = R_0 A^{1/3} \quad [R_0 \approx 1.2\,\text{fm} = 1.2 \times 10^{-15}\,\text{m}], \qquad \rho_{\text{nuclear}} = \frac{M}{\frac{4}{3}\pi R^3} \approx 2.3 \times 10^{17}\,\text{kg m}^{-3} \tag{41}$$

$$\Delta m = [Z m_p + (A - Z) m_n] - M_{\text{nucleus}} \quad \text{[Mass Defect]} \tag{42}$$

$$E_b = \Delta m\, c^2 = \Delta m\ (\text{in u}) \times 931.5\,\text{MeV} \quad \text{[Total Binding} \tag{43}$$

$$\Delta E_{\text{reaction}} = Q = [m_{\text{reactants}} - m_{\text{products}}]c^2 = [BE_{\text{products}} - BE_{\text{reactants}}] \tag{44}$$

Nuclear Traps: Invariance of Nuclear Density

- **Nuclear Density is Constant:** Because $R \propto A^{1/3}$, nuclear volume $V \propto R^3 \propto A$. Thus nuclear density $\rho = \frac{mA}{V} \propto \frac{A}{A} =$ constant. A Uranium nucleus ($A = 238$) has approximately the **same density** as a Helium nucleus ($A = 4$).
- **Radioactivity Excluded by NMC:** Note that α, β, γ decay rate equations and half-life decay laws have been removed from NEET UG under current NMC guidelines; focus strictly on binding energy per nucleon curves, mass defects, and fission/fusion energy calculations.

5 Electronic Devices: Semiconductors (Unit 19)

5.1 Intrinsic & Extrinsic Semiconductors

Carrier Densities, Mass Action Law & Electrical Conductivity

$$n_e n_h = n_i^2 \quad \text{[Law of Mass Action: val} \tag{45}$$

$$n_e = n_h = n_i \quad \text{[Intrinsic Semiconductor]} \tag{46}$$

$$n_e \approx N_d \gg n_h \quad \text{[n-Type Semiconductor:} \tag{47}$$

$$n_h \approx N_a \gg n_e \quad \text{[p-Type Semiconductor:} \tag{48}$$

$$\sigma = e(n_e\mu_e + n_h\mu_h), \qquad I = I_e + I_h = eAE(n_e\mu_e + n_h\mu_h) \quad \text{[Total Drift Current]} \tag{49}$$

5.2 p-n Junction Diodes, Rectifiers & Optoelectronic Devices

Diode Dynamic Resistance & Rectifier Efficiencies

$$r_d = \frac{\Delta V}{\Delta I} \quad \text{[Dynamic Resistance: Low in Forward Bias } (\sim 10 - 100\,\Omega)\text{, High in Reverse Bias } (\sim 10^6\,\Omega)\text{]} \tag{50}$$

$$\eta_{\text{half-wave}} = \frac{P_{\text{dc}}}{P_{\text{ac}}} = \frac{0.406R_L}{r_f + R_L} \implies \eta_{\max} = 40.6\%, \qquad \nu_{\text{ripple}} = \nu_{\text{input}} \tag{51}$$

$$\eta_{\text{full-wave}} = \frac{P_{\text{dc}}}{P_{\text{ac}}} = \frac{0.812R_L}{r_f + R_L} \implies \eta_{\max} = 81.2\%, \qquad \nu_{\text{ripple}} = 2\nu_{\text{input}} \tag{52}$$

5.3 Digital Electronics: Logic Gates & Boolean Algebra

Input A	**Input** B	**AND** $(A \cdot B)$	**OR** $(A + B)$	**NAND** $(\overline{A \cdot B})$	**NOR** $(\overline{A + B})$	**XOR** $(A \oplus B)$	**XNOR** $(\overline{A$
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Table 3: Comprehensive Truth Tables for Fundamental, Universal and Derived Logic Gates.

De Morgan's Theorems & Universal Logic Construction

- **De Morgan's Laws:**
$$\overline{A + B} = \overline{A} \cdot \overline{B}, \qquad \overline{A \cdot B} = \overline{A} + \overline{B}$$
- **Universal Gates Minimal Realizations:**
  - Number of NAND gates to make: NOT = 1, AND = 2, OR = 3, NOR = 4, XOR = 4.
  - Number of NOR gates to make: NOT = 1, OR = 2, AND = 3, NAND = 4, XNOR = 4.

End of Module 04 Formula Handbook (Complete Physics Series)

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